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EDUCATION

New York University

Master of Science - Computer Science

August 2025 - Expected May 2027

Brooklyn, New York

- **Courses:** Design & Analysis of Algorithms, Software Engineering, Machine Learning, Big Data, Operating Systems, Deep Learning, Language models: Reinforcement Learning & Reasoning
- **Teaching Assistant:** GPUs: Architecture and Programming — Grading CUDA programming and GPU profiling labs and assignments for 90 students.

National Institute of Technology, Warangal

Bachelor of Technology - Computer Science Engineering

July 2019 - May 2023

Warangal, Telangana

EXPERIENCE

PulseSync

Software Engineer Intern (AI Harness)

June 2026 - August 2026

Fremont, California

- Built a LangGraph-based **multi-agent AI system** for **medical fraud detection**, designing and evaluating agents across **110+ tools**. Built the microservice using **FastAPI** APIs with **PHI** firewalls to protect patient data.

CloudAngles

Software Engineer (LLMOps & AI Agents)

September 2023 - April 2025

Hyderabad, Telangana

- Built an internal **LLM fine-tuning, quantization and deployment platform** leveraging GPU clusters from AWS and Azure, enabling clients to train models and run inference in-house and avoid third-party data exposure. Applied full fine-tuning and **PEFT (LoRA)** techniques across various model architectures including Llama, Gemma, Mistral, Falcon, and Bloom, and instruct finetuned them on medical, financial and cloud cost datasets to achieve up to **20% accuracy boost** on domain-specific benchmarks.
- Deployed fine-tuned models to production via **vLLM** across multiple GPUs with **GGUF/AWQ** quantization, maintaining a low p95 latency at inference time, and built an **LLM monitoring system** to detect hallucinations, prevent prompt injection, and assess toxicity and sentiment across deployed models in production.
- Delivered **Agentic AI tools** spanning in-house and forward-deployed use cases: built a **RAG application** to query reports sourced from OneDrive and Google Drive, developed a **LangGraph-based agentic flow** for query segregation and action execution using tool connections into warehouse optimization, bin-packing, and truck-management systems spanning **10,000+ SKUs** for a client, and built an **ETL-mapping POC** automating schema mapping via embedding similarity.

4n Digital Solutions

Software Engineer Intern (Application Development)

July 2023 - September 2023

Dubai, U.A.E

- Developed a **RAG-powered forex chatbot** using LangChain, OpenAI API, FAISS, and SerpAPI, with a React frontend, Flask backend, and Firebase Authentication.

TECHNICAL SKILLS

Languages: C, C++, Python, SQL, JavaScript, HTML, CSS, React

Databases & Vector Stores: MySQL, PostgreSQL, OracleSQL, MongoDB, Apache(Spark, Hadoop, Kafka), Pinecone, Milvus, Chroma, PgVector, Neo4j

Frameworks & Libraries: Pytorch, JAX, Transformers, CUDA, Langchain, AutoGen, Pandas, NumPy, Matplotlib, Font Awesome, Bootstrap, xv6

API Development & Testing: Postman, Django, Fast API, Node JS, Express JS, Socket.io

Cloud Services: AWS(EC2, Bedrock, S3, RDS, Lambda, Elastic Beanstalk, Glue, EKS, IAM, Terraform), GCP(Cloud Run, Maps Platform, Vertex AI), HERE Maps Platform

Data Formats: Apache Parquet, Apache Arrow, Microsoft Excel, Google Sheets, CSV, JSON, JSONL

Dev Tools & containers: Git, Travis CI/CD, Argo CD, Linear, Jira, Zenhub, ML Flow, Weights & Biases, Prometheus, Grafana, Nsight, Temporal, Docker, Kubernetes

Certifications: Blockchain Specialization (Coursera), 30 Days of Google Cloud (GDSC NITW)

PROJECTS

Synthetic Robotic Checkers Dataset Generation 📄 🎮 🏠

May 2026

- Built a **VLA data generation pipeline** using **VLLM**-deployed **Gemma 4 (E4B-it)** to control a simulated **Trossen WidowX dual-arm robot** in **MuJoCo**, converting board images into chain-of-thought reasoning and structured JSON actions across **8 low-level primitives**.
- Engineered a **best-of-N validator** sampling **4 candidates/turn** against an **8-check legality gate** and an annealed **3-tier RL opponent curriculum**, cutting illegal/malformed outputs to **near zero** and producing a clean, human-review-free dataset.

Wafer-Scale Top-K k-NN Kernel (Cerebras WSE-2) 🔗

Apr 2026

- Implemented a **CSL kernel** using the Cerebras WSE-2 simulator that shards an **$N \times d$ vector dataset** across a **$P \times P$ PE grid**, using 2D fabric-collective **broadcast/gather** routing and a per-PE **max-heap** for $O(\text{rows} \cdot (d + \log K))$ top-K selection and passed all 6 grader cases including tie-break and duplicate workloads.

xv6 User-Level Threading Library 📺

December 2025

- Built a **user-level N:1 threading library** for xv6 from scratch in C, implementing thread **create/join/yield/exit** on a **cooperative round-robin scheduler** with hand-written **x86 assembly context switching**, plus **mutexes**, **semaphores**, **condition variables**, and **Go-style channels** with FIFO wait queues and direct lock-ownership transfer.
- Applied the library to **producer-consumer** and **writer-priority reader-writer** concurrency patterns, and engineered an **asynchronous I/O subsystem** with a dedicated worker thread and completion channels to prevent blocking syscalls from stalling all threads, demonstrated via a cross-process pipe pipeline.